

Bias, Democracy, and Collaboration: Investigating Neutrality in Wikipedia

CPSC-298 Wikipedia Governance Research Project

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Abstract

Wikipedia has transformed how knowledge is shared and consumed; however, it also raises important questions about neutrality and bias. This research investigates how the collaborative and open nature of Wikipedia reduces or increases bias compared to traditional encyclopedias. Building upon prior research, the study explores sources of bias and develops models to identify biased language at the sentence level.

Additionally, this work examines how Wikipedia reflects democratic practices. While Wikipedia promotes collaboration, small groups of editors often dominate governance. Wikipedia's Talk pages serve as deliberative spaces where politically diverse contributors may improve content quality. This project analyzes Wikipedia data to assess neutrality in discussions, examine consensus-building, study governance of contentious topics, and evaluate participation patterns. Wikipedia thus offers insight into both the possibilities and limitations of democracy in online collaborative knowledge production.

Keywords

Wikipedia, governance, bias detection, online democracy, collaborative systems

1 Introduction

Wikipedia, one of the world's largest collaborative knowledge platforms, operates under an open-editing model that invites anyone to contribute. While this openness has democratized knowledge creation, it also raises complex questions about bias, neutrality, and governance. For example, discussions on politically charged articles such as *Climate Change* or *Gun Politics in the United States* often reveal tensions between consensus-building and ideological polarization among editors.

The central problem this research addresses is how Wikipedia's open, collaborative structure influences the neutrality of its content and the democratic quality of its editorial deliberations. Specifically, we examine whether Wikipedia's design fosters genuinely democratic decision-making—or whether, as prior research suggests, governance tends to consolidate among small groups of experienced editors.

Understanding how Wikipedia handles bias and governance is critical not only for evaluating the reliability of one of the most visited knowledge resources online, but also for informing the design of future collaborative systems. As Wikipedia continues to shape global discourse, identifying how bias emerges, evolves, and

is moderated through collective deliberation offers broader insights into the health of digital democracy.

This paper investigates the following research questions:

- (1) How can we detect and measure biased language in Wikipedia articles at the sentence level?
- (2) How does Wikipedia bias change over time through the revision process?
- (3) To what extent do Wikipedia Talk pages function as democratic spaces of deliberation?
- (4) How do patterns of participation influence consensus-building?
- (5) Does democratic participation reduce bias, or do small groups of dominant editors shape outcomes that increase bias?

The main contributions of this work are:

- A computational framework for detecting and tracking bias in Wikipedia articles using sentence-level machine learning models.
- An empirical analysis of Talk page discussions to assess sentiment, toxicity, and deliberative tone across topics.
- An integrative model connecting editorial diversity and governance patterns with the neutrality of Wikipedia content.
- A prototype system leveraging the Wikipedia API for automated analysis and visualization of editor interactions.

The rest of this paper is organized as follows. Section 2 reviews related work on Wikipedia governance, bias, and online deliberation. Section 3 describes our data sources and analytical methods, including bias detection and governance modeling. Section 4 presents findings from the prototype implementation and sentiment/toxicity analyses. Section 5 interprets the implications of these results for digital democracy and collaborative systems. Section 6 concludes and outlines directions for future work.

2 Related Work

Research on Wikipedia spans multiple domains, including computer science, sociology, communication, and political science. Scholars have examined how Wikipedia governs itself, how collaborative editing impacts content quality, and how bias and neutrality emerge in open knowledge systems. This section organizes prior research into three major themes: governance and community culture, bias and neutrality, and democratic deliberation.

2.1 Wikipedia Governance and Community Culture

Reagle’s seminal work on Wikipedia culture [5] explores the norms of collaboration and consensus that underpin the platform’s community ethos, often summarized by the term “good faith collaboration.” Konieczny [4] and Shaw and Hill [6] expanded on this by identifying governance patterns that resemble oligarchic structures, where small, highly active groups control decision-making despite the open nature of the platform. Studies of editorial hierarchies show that informal authority and technical expertise influence whose edits persist and whose are reverted, shaping both policy enforcement and content stability.

2.2 Bias and Neutrality in Wikipedia

A second line of research examines whether Wikipedia’s open editing model fosters or mitigates bias. Hube [2] developed computational approaches to detect biased phrasing and framing in article text, showing that even factual statements can carry subtle ideological leanings. Greenstein and Zhu [1] compared political articles and found evidence of ideological balance emerging over time due to countervailing edits by users with opposing viewpoints. These studies underscore that bias in Wikipedia is dynamic—it evolves through ongoing revision and editorial negotiation.

2.3 Democratic Deliberation and Online Collaboration

Talk pages serve as deliberative spaces where editors debate sources, tone, and factual claims. Klemp and Forcehimes [3] highlight how these discussions mirror the ideals of deliberative democracy: open participation, reasoned argument, and consensus-seeking. Shi et al. [7] empirically demonstrated that politically diverse participation improves article quality and reduces ideological slant. However, participation remains uneven, and contentious topics often reveal asymmetrical power dynamics where a few editors dominate debates.

2.4 Our Work in Context

By combining machine learning–based bias detection with sentiment and toxicity analysis of Talk pages, this work investigates whether inclusive, democratic participation correlates with greater neutrality in Wikipedia’s content. This integrated perspective helps fill a key gap between computational text analysis and sociotechnical studies of online governance.

3 Methodology

This study employs a mixed-method computational approach that bridges our *motivating research questions*—concerning bias, neutrality over time, and democratic deliberation on Wikipedia—with the *operationalized questions* that can be answered using observable Wikipedia data. Because constructs such as “bias,” “deliberation,” and “democratic participation” cannot be measured directly, our methodology centers on identifying appropriate *proxies* and applying machine learning tools to compare these signals across multiple Wikipedia Talk pages.

All data-collection code, processing scripts, and instructions for reproducing every figure in this paper are available in our GitHub repository: github.com/brookeengland/Wikipedia-Bias-and-Democracy.

3.1 Operationalizing Research Questions Through Proxies

Our motivating questions ask how Wikipedia handles bias, how discussions evolve, and whether Talk pages reflect democratic deliberation. To answer these using measurable data, we identify the following proxies:

- **Bias in content:** sentiment cues, lexical framing, and toxicity patterns (following Hube’s sentence-level bias indicators).
- **Deliberative quality:** proportion of Neutral vs. Positive/Negative comments.
- **Adversarial or harmful discourse:** toxicity levels (Low/Medium/High).
- **Democratic participation:** number of unique contributors and balance of participation across editors.

These proxies are commonly used in prior research on Wikipedia governance and online deliberation, and they provide computable indicators aligned with our research questions.

3.2 Data Collection

Data were collected exclusively from Wikipedia Talk pages through the Wikipedia API using the `action=query` and `prop=revisions` endpoints. For each topic, we retrieved the full Talk page content and extracted individual comments using a timestamp-based regular expression matching the standard Wikipedia signature format.

Our updated analysis script supports the comparison of **multiple Talk pages at once**, allowing cross-topic evaluation (e.g., *United States*, *Climate change*, *Elon Musk*).

3.3 Data Processing

Each Talk page undergoes the following preprocessing steps:

- (1) **Comment segmentation:** isolate user comments based on signature patterns.
- (2) **Cleaning:** remove usernames, timestamps, templates, and markup.
- (3) **Chunking:** split long comments into sections that fit within model token limits.
- (4) **Normalization:** store sentiment and toxicity results per-topic to enable direct comparison.

Only the textual content of comments was processed, as non-textual metadata does not contribute to the deliberative or tonal characteristics of discussions.

3.4 Analysis Methods

We analyze each extracted comment using two transformer-based machine learning models:

- **Sentiment Analysis:** Using the TabularAI multilingual model, comments are categorized as Positive, Neutral, or Negative. Neutral sentiment is treated as a proxy for reasoned, deliberative communication.
- **Toxicity Detection:** Using the Unitary Toxic-BERT model, each comment is assigned a toxicity level (Low, Medium, High).

For each Talk page, we aggregate:

- sentiment label distributions,
- sentiment confidence levels,
- toxicity levels,
- stacked sentiment percentages.

The updated script automatically generates comparative visualizations across all selected topics, allowing us to operationalize questions such as: (1) whether politically contentious topics show higher toxicity, (2) whether some pages exhibit more democratic participation than others, and (3) how the tone of discourse varies by subject matter.

All machine learning analysis and visualization code is included in our repository and can be run with a single Python script.

3.5 Reproducibility

To ensure reproducibility, the repository contains:

- full API querying code,
- preprocessing and comment-extraction scripts,
- instructions for installing Python dependencies,
- the full multi-topic analysis script used to generate the results in this paper.

Not all implementation details appear in the PDF; however, the GitHub repository provides everything needed to regenerate our results exactly.

3.6 Ethical Considerations

All analyzed data were collected from publicly available Wikipedia Talk pages in accordance with [Wikimedia’s Terms of Use](#). No private or user-identifiable information was collected or stored. The study complies with institutional guidelines for research using publicly accessible online data.

4 Results

This section presents the outcomes of the bias detection and Talk page tone analyses, focusing on variation across topics and the relationship between participation diversity and neutrality.

4.1 Talk Page Sentiment and Toxicity

Sentiment analysis showed that discussions typically remained neutral, with only slight negative comments detected.

The automated talk-page analysis produced per-topic summaries (counts and label distributions); the full machine-readable results are available in `paper/article_comparisons_results.json` and per-subset JSON files in the same directory. Representative comparison figures were saved to `figures/article_comparisons_default.png` (and other subset-specific PNGs).

Table 1 summarizes the default subset (topics used in the main analysis). Percentages are relative to the number of comments extracted per talk page.

These automated summaries show that the majority of extracted comments were labeled “Neutral” by the sentiment classifier, and toxicity scores were low for the sampled comments across topics. The saved figures (e.g., `figures/article_comparisons_default.png`) visualize the label distributions and confidence levels used to generate Table 1.

Topic	# Comments	Neutral (%)	Negative (%)
Donald Trump	14	92.9	7.1
Antisemitism	2	100.0	0.0
Capitalism	77	94.8	5.2
Atheism	4	100.0	0.0
British National Party	3	100.0	0.0
Feminism	4	100.0	0.0
Gentrification	2	100.0	0.0

Table 1: Summary of Comments by Topic

Notes and limitations: the counts reflect the subset of comments the script could extract (see the JSON files for exact counts), the sentiment model is a general-purpose classifier and may under- or over-count valenced language in politically charged discussions, and toxicity estimates are coarse (“Low/Medium/High”) based on classifier score thresholds. We treat these results as descriptive and exploratory; interpretation and causal claims are deferred to the Discussion.

4.2 Subset summaries

The analysis also produced subset-specific summaries (“political”, “social”, and “extremes”) whose JSON outputs are available as `paper/article_comparisons_results.political.json`, `paper/article_comparisons_results.social.json`, and `paper/article_comparisons_results.extremes.json`. In brief:

- Political subset (Donald Trump, British National Party, Capitalism): distributions remain dominated by Neutral labels; Capitalism contributed the largest comment sample (77 comments) and therefore drives the aggregated counts.
- Social subset (Feminism, Gentrification, Atheism): small sample sizes (2–4 comments per topic) but similarly high proportions of Neutral labels in the extracted comments.
- Extremes subset (British National Party, Antisemitism): very small samples (2–3 comments per topic) and therefore highly variable; interpret cautiously.

5 Discussion

5.1 Interpretation of Results

The sentiment and toxicity analyses show that most Wikipedia Talk page discussions in our dataset were labeled as Neutral, even for topics that are commonly viewed as political or controversial. This suggests that editors generally communicate in a measured way and follow Wikipedia’s expectations for civil discussion. Toxicity levels were also consistently low, which further supports the idea that the platform’s community guidelines help shape constructive dialogue.

At the same time, the results vary depending on the number of comments available for each topic. Pages with many comments, such as *Capitalism*, provide stronger evidence of stable and neutral discussion patterns. In contrast, topics with only a few comments (for example, *Gentrification* or *Atheism*) do not offer enough data to draw firm conclusions. In these cases, high neutrality percentages may simply reflect limited participation rather than the presence of consensus.

Overall, our findings suggest that while the tone of Talk page discussions tends to be neutral, this does not automatically mean that the underlying decision-making processes are broad or representative.

5.2 Implications

These findings have two main implications for understanding Wikipedia as a collaborative platform. First, the neutral tone of discussions suggests that Wikipedia's policies, such as assuming good faith and working toward consensus, appear to influence how editors interact. This is encouraging for the quality of article content, as constructive discussion is often linked to more accurate and balanced writing.

Second, however, neutrality in tone does not guarantee democratic participation. Several Talk pages were shaped by only a few editors, which means that decision-making may still be concentrated among a small group. This aligns with existing research showing that online communities can develop informal hierarchies over time. Therefore, it is important to think about both *how* editors communicate and *who* participates when evaluating Wikipedia's governance.

For broader digital platforms, these patterns highlight the challenge of supporting both civil communication and inclusive participation. A platform may succeed at encouraging respectful dialogue but still struggle to involve a diverse group of contributors.

5.3 Limitations

This study has several limitations. The number of comments varied greatly from topic to topic, which affects how much we can infer from each case. Some pages had only two or three comments, while others had dozens. The sentiment and toxicity models we used are also general-purpose tools, which means they might miss subtleties in Wikipedia conversations, such as sarcasm or complex arguments.

Additionally, our analysis focused only on English-language Wikipedia and a limited set of topics. Other language communities may have different norms, participation patterns, or conflict dynamics. Finally, while we measured the number of contributors, we did not study their roles or influence in detail. A deeper analysis of editor networks would help clarify how decisions are made.

5.4 Future Work

Future work could expand both the methods and the scope of this study. More advanced language models could help capture context and detect subtle forms of bias more reliably. Another useful direction would be a temporal analysis that follows discussions over time to see how tone or toxicity shifts during major events or controversies.

Additional research could also examine how editors interact with one another by mapping participation networks, which would help reveal whether certain contributors consistently guide discussions. Comparing different language Wikipedias would provide insight into how culture and community norms shape governance. Finally, connecting Talk page discussions directly to changes in article content would show more clearly how deliberation influences Wikipedia's final output.

6 Conclusion

This study investigated how Wikipedia's collaborative governance shapes neutrality in Talk page discussions. Using sentiment and toxicity analysis across multiple topics, we found that most comments were classified as Neutral and showed low toxicity, suggesting that Wikipedia's norms help support constructive dialogue even around controversial subjects.

At the same time, many Talk pages were driven by only a few active editors, indicating that participation is not always broadly distributed. This raises questions about how democratic Wikipedia's deliberation processes truly are, despite the platform's open nature.

Our work provides a simple framework for evaluating discourse tone and participation patterns using machine learning tools. Future work should examine editor networks, expand to other languages, and connect Talk page discussions to the actual edits made to articles. Understanding these dynamics is important for assessing how online collaborative systems balance openness, neutrality, and shared decision-making.

Acknowledgments

We thank our peers and instructors in CPSC-298 for their feedback and support during this research project.

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A AI Usage Documentation

A.1 Literature Review

We used Claude (Anthropic) to assist with organizing and synthesizing the initial literature review. The AI helped identify thematic groupings across multiple research papers on Wikipedia governance, bias detection, and online deliberation. Specifically, we provided Claude with paper titles, abstracts, and key excerpts, then asked it to:

- Identify recurring themes and concepts across the literature
- Suggest organizational structures for the Related Work section
- Generate preliminary summaries of research methodologies and findings

All AI-generated summaries were subsequently verified by reading the complete original papers. The final Related Work section

(Section 2) was written entirely by the human authors after reviewing primary sources. AI assistance served only as an organizational scaffold for understanding the research landscape.

A.2 Data Analysis

We employed two pre-trained transformer-based machine learning models to analyze Wikipedia Talk page comments:

- **Sentiment Analysis:** The TabularisAI multilingual sentiment classification model categorized each comment as Positive, Neutral, or Negative. This provided a proxy measure for deliberative tone.
- **Toxicity Detection:** The Unitary Toxic-BERT model assigned toxicity levels (Low, Medium, High) to assess the presence of harmful or adversarial discourse.

These models were applied programmatically through our Python analysis pipeline without manual intervention in the classification process. Model outputs were treated as quantitative indicators rather than ground truth, and all interpretations of model results were conducted by the human authors.

Additionally, we used Claude to assist with debugging data processing scripts, particularly for:

- Troubleshooting regular expression patterns for comment extraction
- Optimizing JSON data structure for cross-topic comparisons
- Identifying edge cases in timestamp parsing

A.3 Writing Assistance

Claude was used to improve clarity, concision, and flow in several sections of this paper:

- **Abstract:** AI suggested more concise phrasing for research contributions while preserving technical accuracy
- **Methodology:** AI helped reformulate complex technical descriptions for better readability
- **Discussion:** AI identified areas where transitions between paragraphs could be strengthened

All substantive arguments, interpretations, and research conclusions were developed entirely by the human authors. AI suggestions were selectively incorporated only when they improved readability without altering meaning or introducing unsupported claims.

A.4 Code Development

AI assistance (Claude) was used extensively in developing the data collection and analysis pipeline:

- Generating boilerplate code for Wikipedia API queries using the `action=query` and `prop=revisions` endpoints
- Writing regular expressions to extract user comments based on Wikipedia signature format
- Debugging preprocessing functions for cleaning markup and templates
- Structuring comparative visualization code using `matplotlib`
- Creating the multi-topic analysis framework that generates per-topic JSON outputs

All AI-generated code was reviewed line-by-line, tested on sample data, and modified by the human authors before being applied to the full dataset. The final implementation is available in our

GitHub repository (<https://github.com/brookeengland/Wikipedia-Bias-and-Democracy>) with full documentation.

A.5 Verification

We implemented rigorous verification procedures for all AI-assisted content:

- (1) **Literature claims:** Every factual statement about prior research was verified against the original papers. All citations were checked for accuracy, and direct quotations were confirmed word-for-word.
- (2) **Technical descriptions:** All methodology descriptions were cross-referenced with official model documentation (TabularisAI, Toxic-BERT) and Wikipedia API specifications.
- (3) **Code validation:** All scripts were manually reviewed for logical correctness and tested on sample data before being applied to the full dataset. Output distributions were sanity-checked against expected ranges.
- (4) **Data interpretation:** The Results and Discussion sections were written entirely by human authors without AI generation. All interpretations were grounded in the observed data and supported by prior literature.
- (5) **Reproducibility:** All code and data processing steps were documented in the GitHub repository to ensure independent verification of our findings.

A.6 Limitations of AI Assistance

For transparency, we explicitly note that AI tools were **not** used for:

- Formulating the research questions or hypotheses
- Designing the overall study methodology or analytical framework
- Selecting which topics to analyze or which proxies to measure
- Interpreting statistical results or drawing conclusions
- Making editorial decisions about which findings to emphasize
- Generating novel theoretical insights or arguments

All core intellectual contributions—including research design, proxy selection, model interpretation, and synthesis of findings—originated with and were validated by the human authors.

A.7 Ethical Reflection

We acknowledge that the use of AI in academic research raises important questions about authorship, originality, and intellectual rigor. Our approach prioritizes transparency through comprehensive documentation of all AI usage. We maintain that human judgment and domain expertise remain central to the research process, with AI tools serving as assistants rather than autonomous agents.

By documenting the specific ways AI contributed to this work, we enable readers to evaluate the role of automation in our findings and to distinguish between computational processing, AI-assisted editing, and human interpretation. This transparency is essential for maintaining academic integrity while responsibly leveraging emerging technologies.